

MATH 2101 LINEAR ALGEBRA I-SOLUTIONS TO SOME PRACTICE PROBLEMS

1. VECTOR SPACES

- (6) Let $v_1, \dots, v_r$ be a set of linearly independent vectors in a vector space V . Suppose $v' \notin \text{span}(\{v_1, \dots, v_r\})$. Then $\{v_1, \dots, v_r, v'\}$ is a set of linearly independent vectors.

Sol: Consider the equation:

$$(*) \quad a_1 v_1 + \dots + a_r v_r + a' v' = 0.$$

Suppose $a' \neq 0$. Then

$$v' = \frac{a_1}{-a'} v_1 + \dots + \frac{a_r}{-a'} v_r$$

This contradicts that $v' \notin \text{span}(\{v_1, \dots, v_r\})$.

Hence, we must have that $a' = 0$. In such case,

$$a_1 v_1 + \dots + a_r v_r = 0.$$

Then the linearly independence of $\{v_1, \dots, v_r\}$ implies that $a_1 = \dots = a_r = 0$. In other words, $a_1 = \dots = a_r = a' = 0$ is the only solution for $(*)$. Hence, $v_1, \dots, v_r, v'$ form a set of linearly independent vectors.

Remark 1.1. In general, one can apply the above exercise multiple times to obtain the general fact that: Suppose $\{v_1, \dots, v_i\}$ is a set of linearly independent vectors in a vector space V . Then one can find vectors $w_1, \dots, w_s$ such that $\{v_1, \dots, v_i, w_1, \dots, w_s\}$ forms a basis for V .

- (7) Use the previous question to find a basis for $\mathbb{R}^3$ that contains the vector $(1 \ 2 \ -1)^T$.

Sol: Let $v = (1 \ 2 \ -1)^T$. Let $w_1 = (1 \ 0 \ 0)^T$. Since w_1 is not a scalar multiple of $(1 \ 2 \ -1)^T$, $\{w_1, v\}$ is linearly independent. Let $w_2 = (0 \ 1 \ 0)^T$. Note that there is no solution for the equation:

$$\begin{pmatrix} 1 & 1 \\ 2 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}.$$

Hence, $(0 \ 1 \ 0)^T$ is not in $\text{span}(\{v, w_1\})$. By the previous exercise, $\{v, w_1, w_2\}$ is still linearly independent. Hence, since $\dim(\mathbb{R}^3) = 3$, $\{v, w_1, w_2\}$ forms a basis for $\mathbb{R}^3$.

2. MATRIX REPRESENTATIONS

- (1) Let $P_2(\mathbb{R})$ be the vector space of all polynomials of degree less than or equal to 2 (with coefficients in $\mathbb{R}$). Let $\beta = \{1, x, x^2\}$ and let $\beta' = \{1 + x, x + x^2, 1 + x^2\}$ be two ordered bases for $P_2(\mathbb{R})$. Let $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be the map of differentiation i.e. $T(f) = \frac{df}{dx}$. Find the matrix representation $[T]_{\beta}^{\beta'}$. Is T invertible?

Sol:

$$T(1) = 0 = 0(1 + x) + 0(x + x^2) + 0(1 + x^2).$$

$$T(x) = 1 = \frac{1}{2}(1 + x) - \frac{1}{2}(x + x^2) + \frac{1}{2}(1 + x^2)$$

(solve an equation if you do not observe the expression!)

$$T(x^2) = 2x = (1)(1 + x) + (1)(x + x^2) + (-1)(1 + x^2).$$

The matrix representation $[T]_{\beta}^{\beta'}$ is:

$$\begin{pmatrix} 0 & \frac{1}{2} & 1 \\ 0 & -\frac{1}{2} & 1 \\ 0 & \frac{1}{2} & -1 \end{pmatrix}.$$

Since the determinant of the matrix is zero, the matrix is not invertible. Hence T is not invertible.

- (2) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by

$$T(x_1, x_2, x_3) = (x_1 + x_2 - x_3, x_2 + x_3, -x_1 - x_2 - x_3).$$

Is T invertible? If so, find T^{-1} .

Sol: Let β be the standard basis for $\mathbb{R}^3$. The standard matrix representation of T is

$$[T]_{\beta} = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ -1 & -1 & -1 \end{pmatrix}.$$

Let P be the above matrix representation. Then

$$\det(P) = \det \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} - \det \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = -2.$$

Hence, P is invertible and so T is also invertible.

We also have (skip the computations):

$$P^{-1} = \begin{pmatrix} 0 & -1 & -1 \\ \frac{1}{2} & 1 & \frac{1}{2} \\ -\frac{1}{2} & 0 & -\frac{1}{2} \end{pmatrix}.$$

Then $T^{-1}(x_1, x_2, x_3) = (-x_2 - x_3, \frac{1}{2}x_1 + x_2 + \frac{1}{2}x_3, -\frac{1}{2}x_1 - \frac{1}{2}x_3)$.

- (3) Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ given by:

$$T(x_1, x_2, x_3) = (x_1 + 3x_2 - x_3, x_1 + 2x_3, x_2 + 4x_3)$$

and let $S : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ given by:

$$S(x_1, x_2, x_3) = (x_2 + x_3, x_1 + x_3, x_1 + x_2, x_1 + x_2 + x_3).$$

Find $(S \circ T)(x_1, x_2, x_3)$.

Sol: Let β and β' be the standard basis for $\mathbb{R}^3$ and $\mathbb{R}^4$ respectively. The standard matrix representation of T is

$$[T]_{\beta} = \begin{pmatrix} 1 & 3 & -1 \\ 1 & 0 & 2 \\ 0 & 1 & 4 \end{pmatrix}$$

and the standard matrix representation of S is

$$[S]_{\beta'}^{\beta} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

Then

$$(2.1) \quad [S \circ T]_{\beta}^{\beta'} = [S]_{\beta'}^{\beta} [T]_{\beta} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 & -1 \\ 1 & 0 & 2 \\ 0 & 1 & 4 \end{pmatrix}$$

$$(2.2) \quad = \begin{pmatrix} 1 & 1 & 6 \\ 1 & 4 & 3 \\ 2 & 3 & 1 \\ 2 & 4 & 5 \end{pmatrix}.$$

Thus

$$(S \circ T)(x_1, x_2, x_3) = (x_1 + x_2 + 6x_3, x_1 + 4x_2 + 3x_3, 2x_1 + 3x_2 + x_3, 2x_1 + 4x_2 + 5x_3).$$